中国海洋大学海德学院

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Statistical Modelling and Inference

Week 5, Lecture 1

Outline:

i. Least Squares Estimation

ii. Scatter Plots and Correlation

iii. Coefficient of Determination

Review of Regression Analysis

- Regression analysis is used to:
  - Explain the **impact** of changes in an independent variable on the dependent variable
  - **Predict** the value of a dependent variable based on the value of at least one independent variable

Dependent variable: the variable we wish to study

Independent variable: the variable used to study the dependent variable

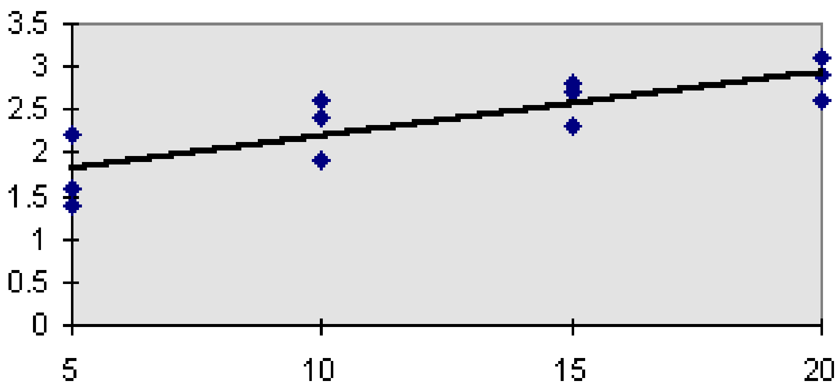
Simple Linear Regression Model

- Only **one** independent variable, x
- Relationship between x and y is described by a linear function
- Changes in y are assumed to be caused by changes in x

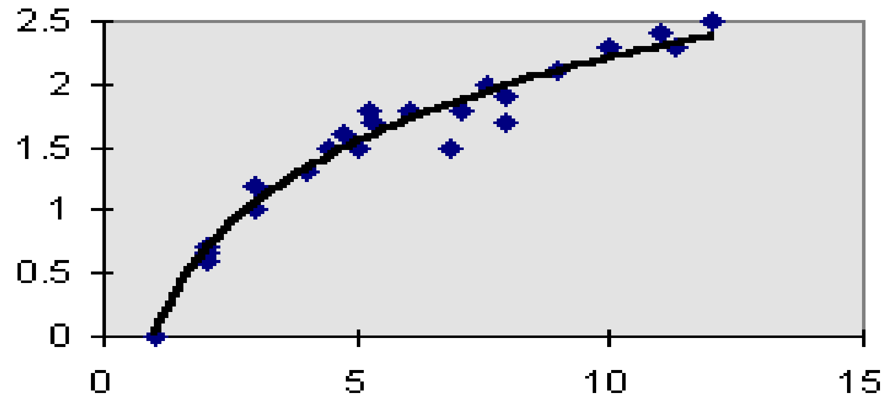
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Types of Regression Models

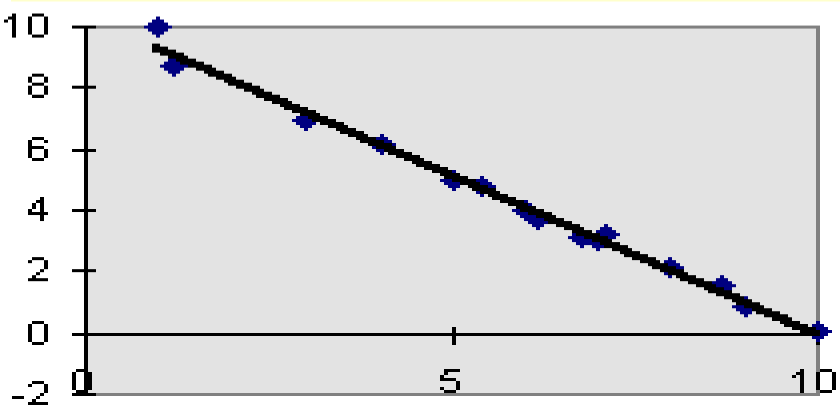
Positive Linear Relationship



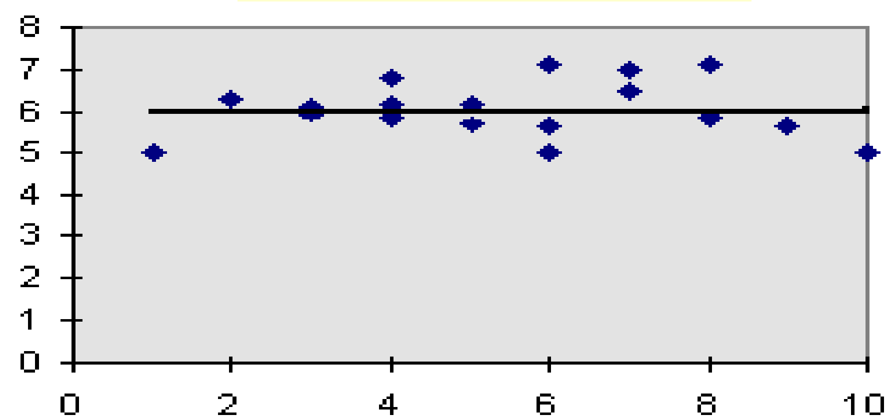
Relationship NOT Linear



Negative Linear Relationship



No Relationship



Linear Regression Assumptions

- Error values (ε) are statistically independent
- Error values are normally distributed for any given value of x
- The probability distribution of the errors is normal
- The probability distribution of the errors has constant variance
- The underlying relationship between the x variable and the y variable is linear

Population Linear Regression

The population regression model:

Population y intercept

Population Slope Coefficient

Independent Variable

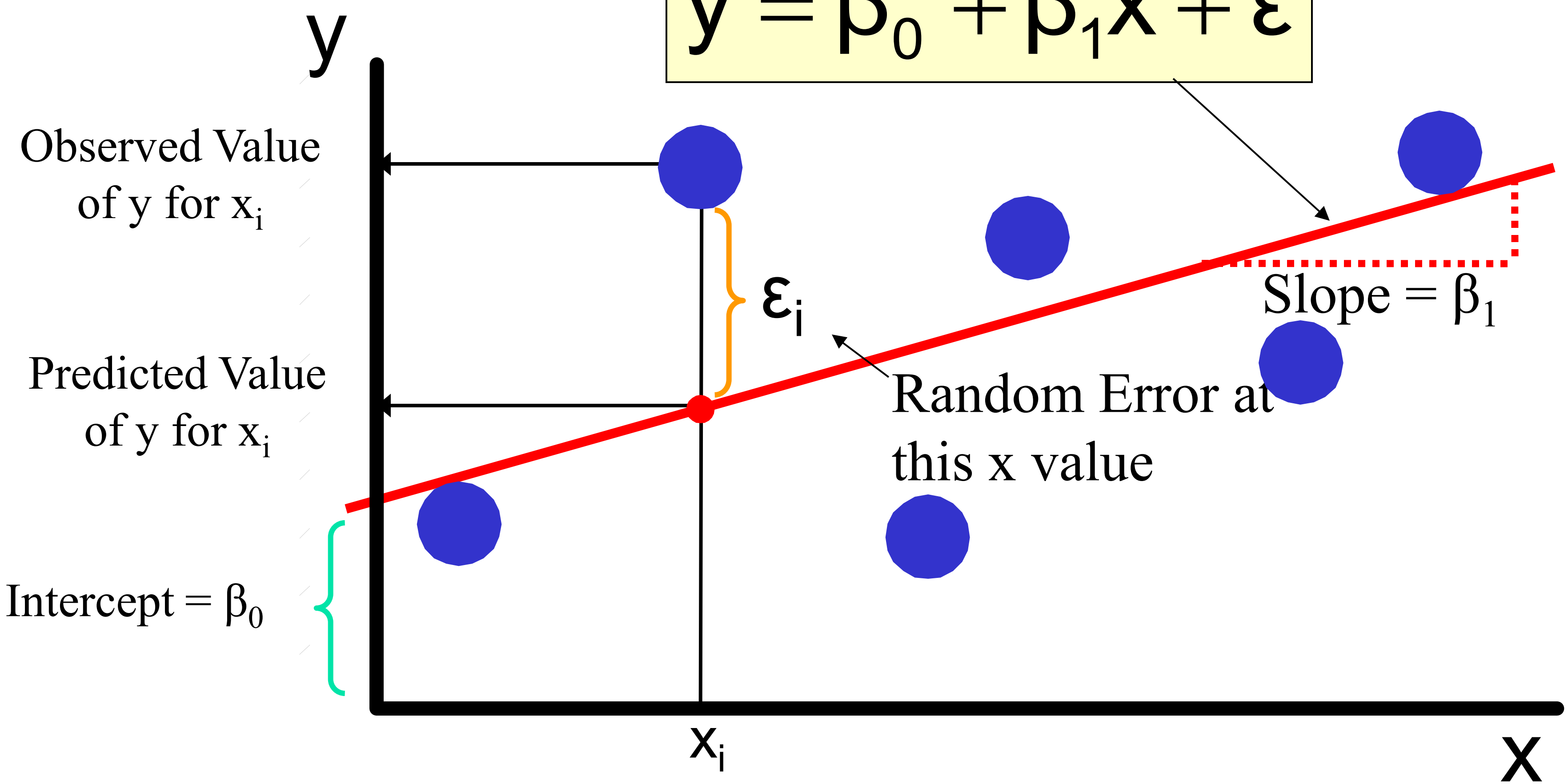
Random Error term, or residual

Dependent Variable

$$y = \underbrace{\beta_0 + \beta_1 x}_{\text{Linear component}} + \underbrace{\varepsilon}_{\text{Random Error component}}$$

Population Linear Regression

$$y = \beta_0 + \beta_1 x + \varepsilon$$



Estimated Regression Model

The sample regression line provides an estimate of the population regression line

Estimated or predicted value of y

Estimate of the regression intercept

Estimate of the regression slope

Independent variable

$$\hat{Y}_{\mathrm{i}} = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

The individual random error terms e_i have a mean of zero

Least Squares Criterion

- β_0 and β_1 are obtained by finding the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of the squared residuals.

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2$$

The Least Squares Estimates

The formulas for Slope is $\hat{\beta}_1$ and intercept $\hat{\beta}_0$ are:

$$\hat{\beta}_1 = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

or,

$$\hat{\beta}_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

Complete derivation on board

Least Squares Regression Properties

- The sum of the residuals from the least squares regression line is 0.

$$\sum_{i=1}^{n} \left(Y - \hat{Y}\right) = 0$$

- The sum of the squared residuals is a minimum (minimized)

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y - \hat{Y}\right)^2$$

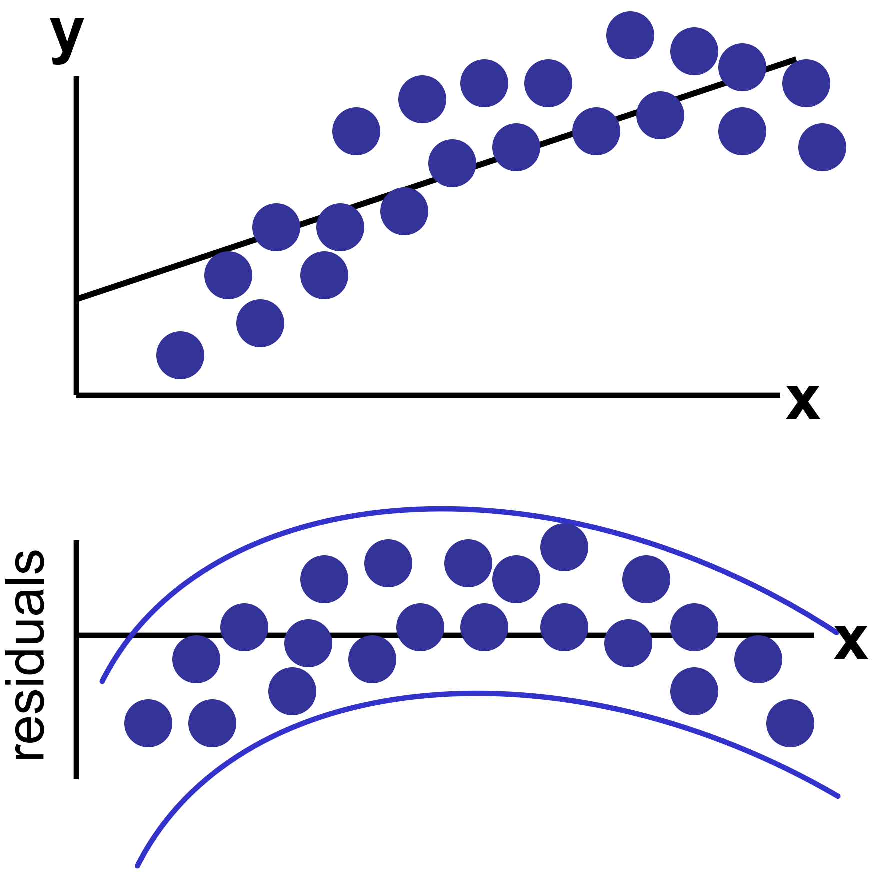
- The simple regression line always passes through the mean of the y variable and the mean of the x variable

- The least squares coefficients are unbiased estimates of β_0 and β_1

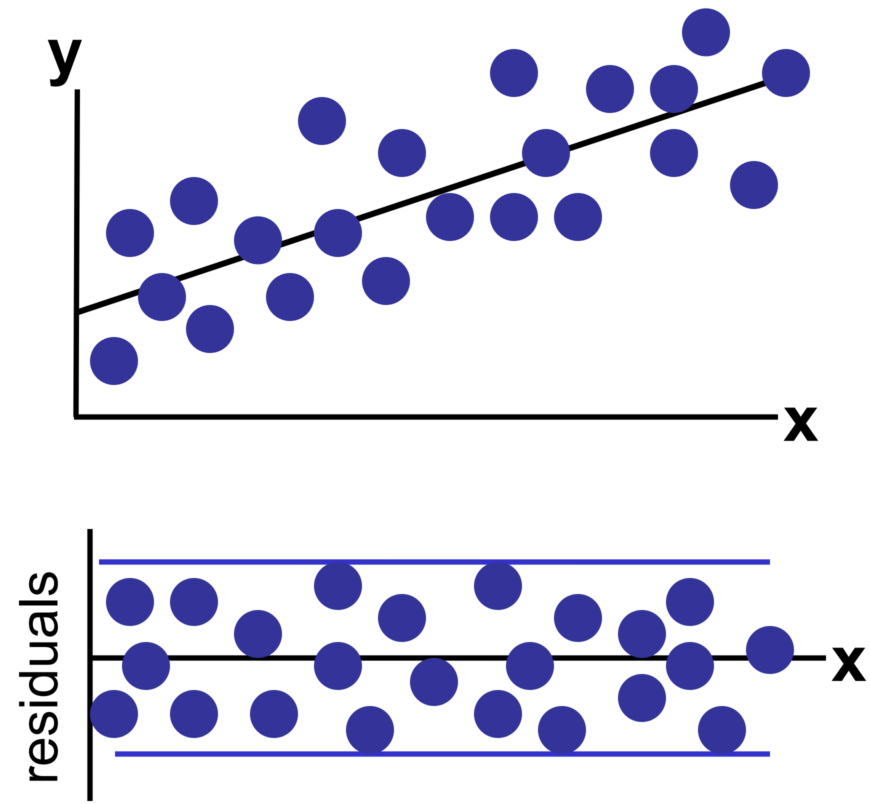
Residual Analysis

- Purposes
  - Examine for linearity assumption
  - Examine for constant variance for all levels of x
  - Evaluate normal distribution assumption
- Graphical Analysis of Residuals
  - Can plot residuals vs. x
  - Can create histogram of residuals to check for normality

Residual Analysis for Linearity

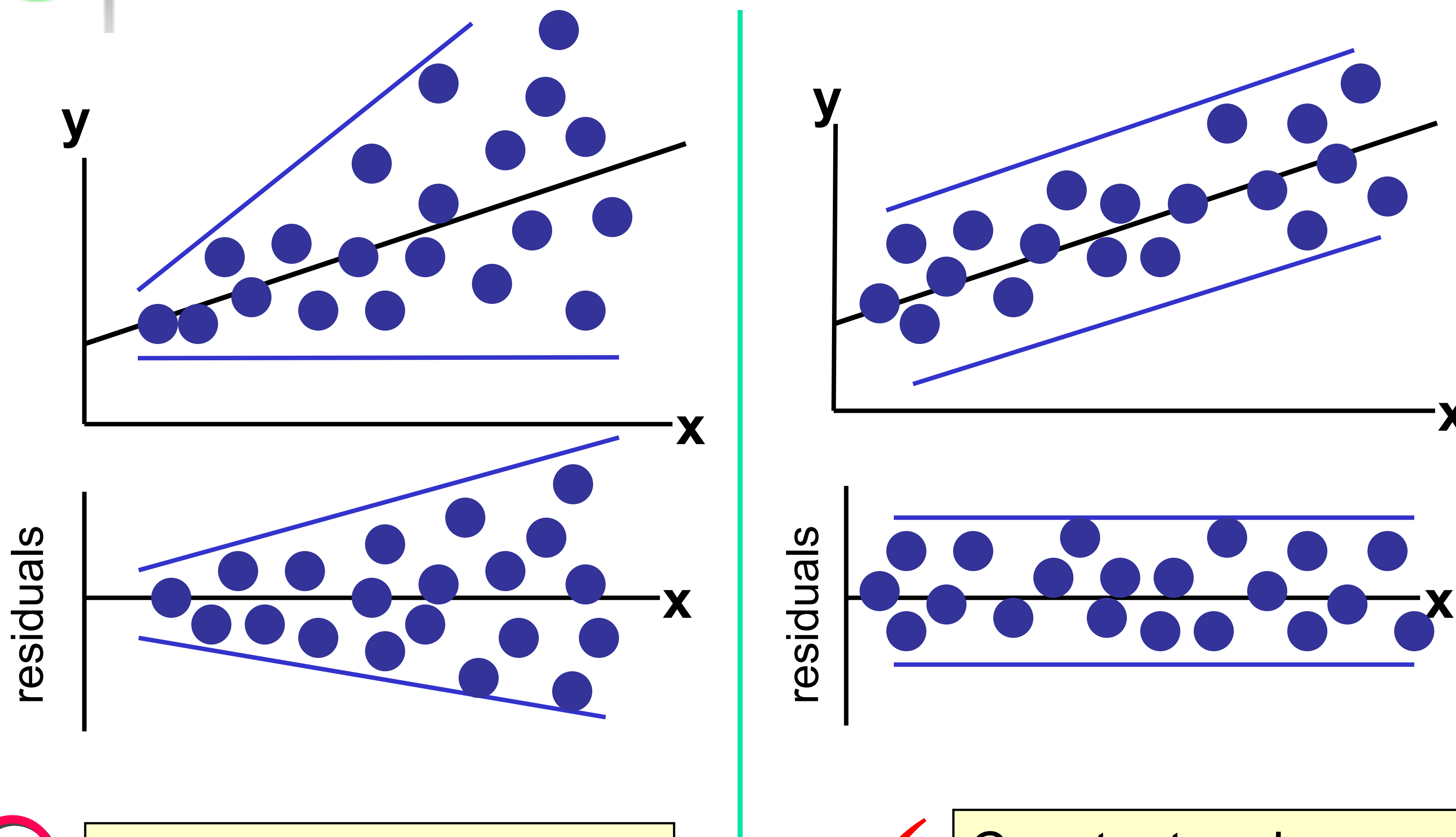


Not Linear



Linear

Residual Analysis for Constant Variance

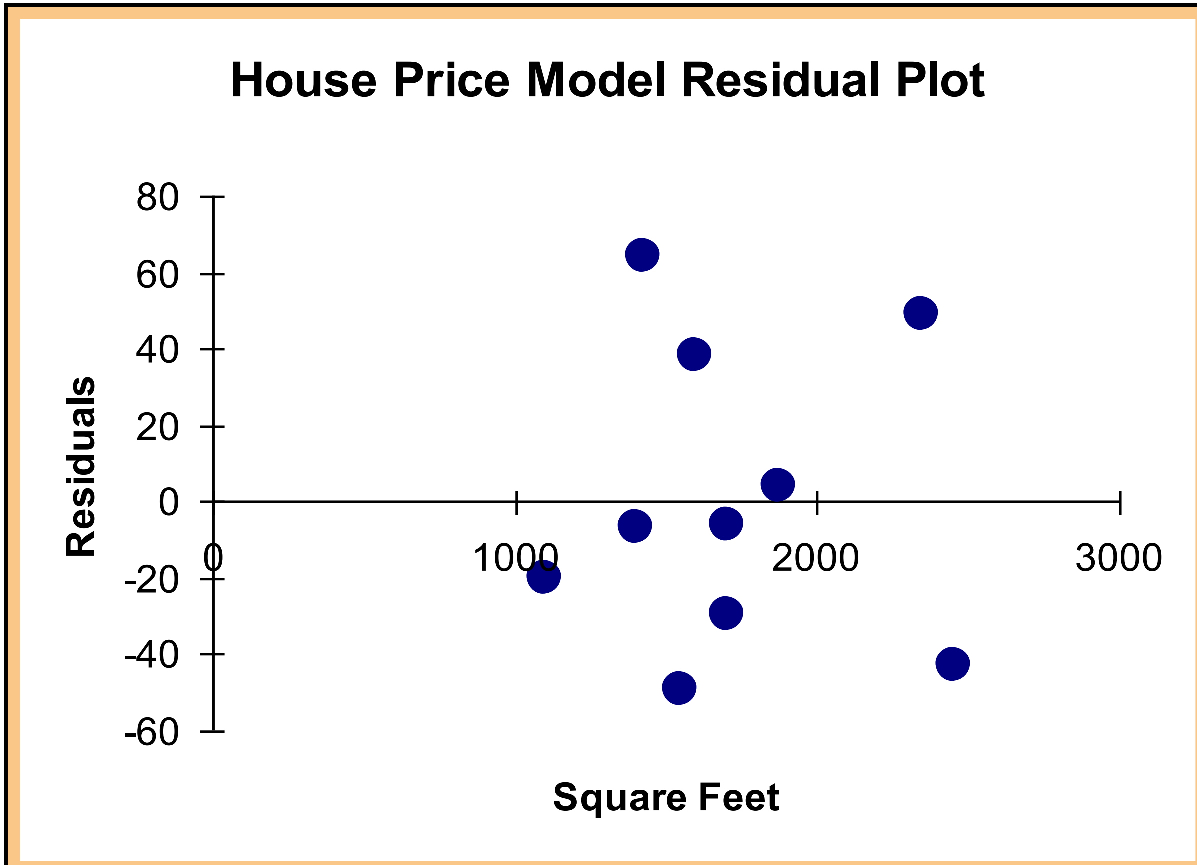


Non-constant variance

Constant variance

Excel Output

RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.92316	-6.923162
2	273.87671	38.12329
3	284.85348	-5.853484
4	304.06284	3.937162
5	218.99284	-19.99284
6	268.38832	-49.38832
7	356.20251	48.79749
8	367.17929	-43.17929
9	254.6674	64.33264
10	284.85348	-29.85348

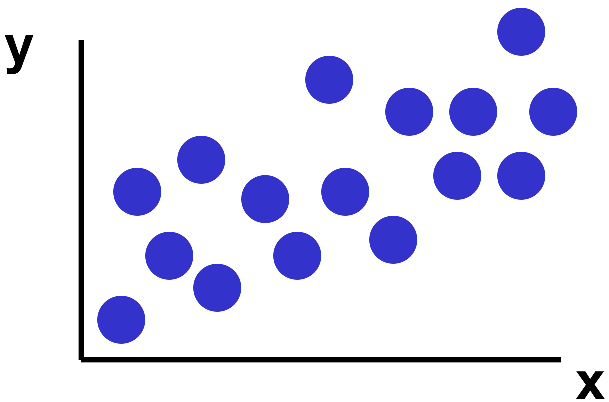


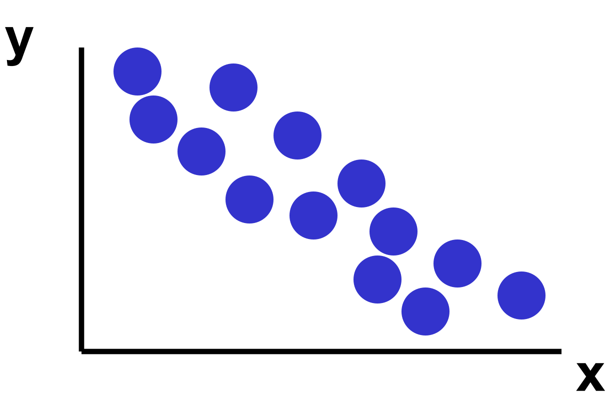
Scatter Plots and Correlation

- A scatter plot (or scatter diagram) is used to show the relationship between two variables
- Correlation analysis is used to measure strength of the association (linear relationship) between two variables
  - Only concerned with strength of the relationship
  - No causal effect is implied

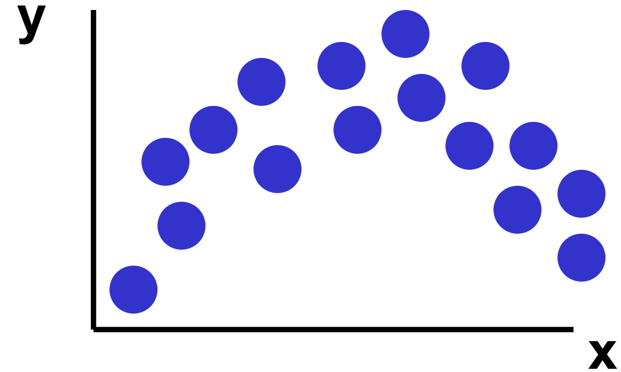
Scatter Plot Examples

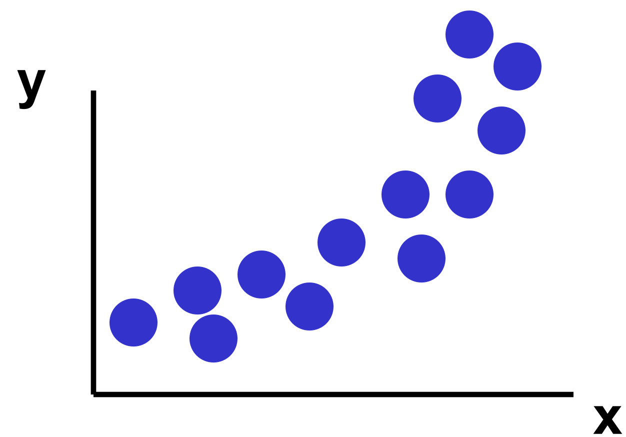
Linear relationships





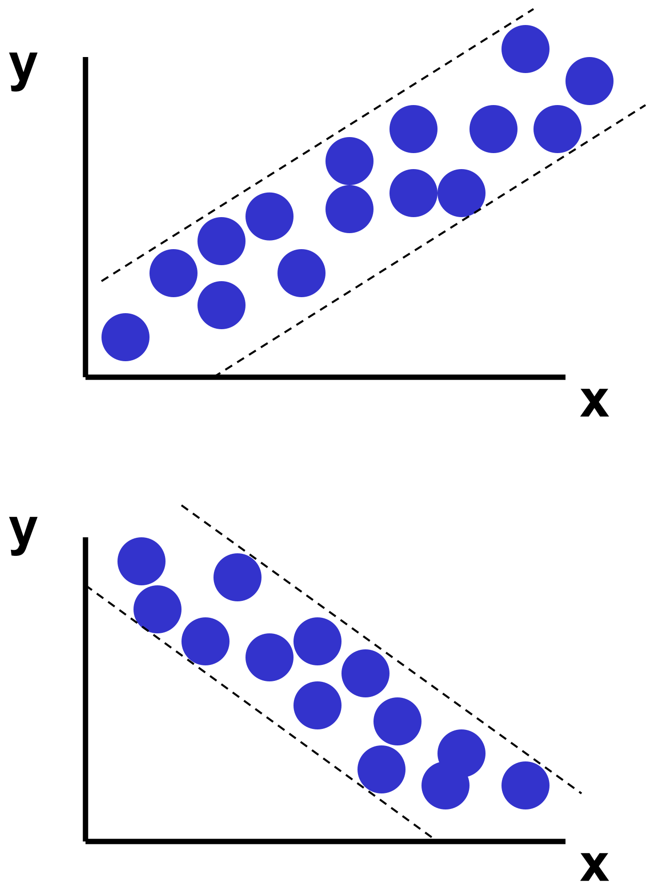
Non-linear relationships



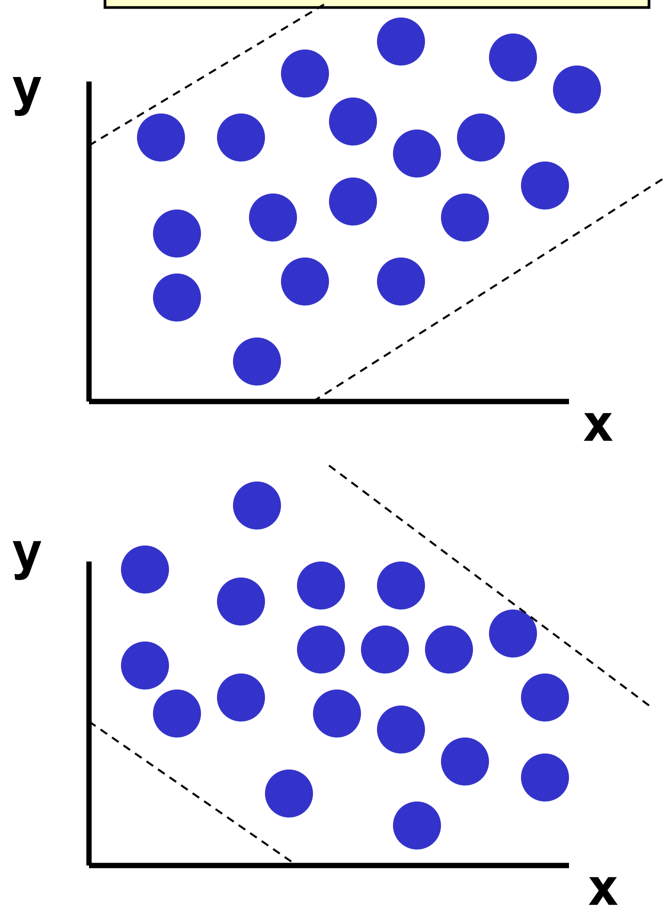


Scatter Plot Examples

Strong relationships

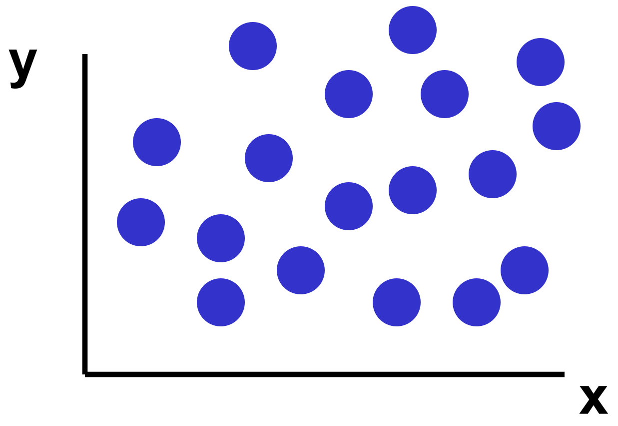


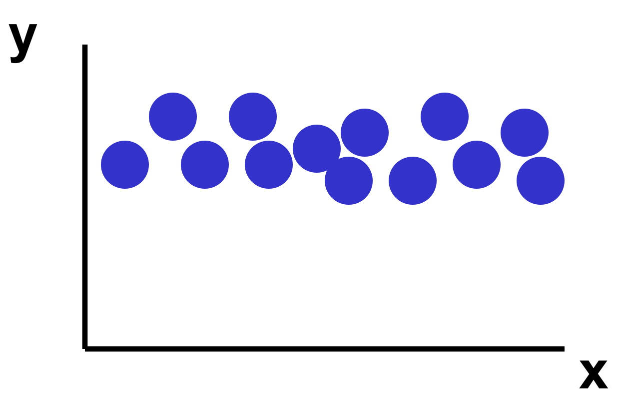
Weak relationships



Scatter Plot Examples

No relationship





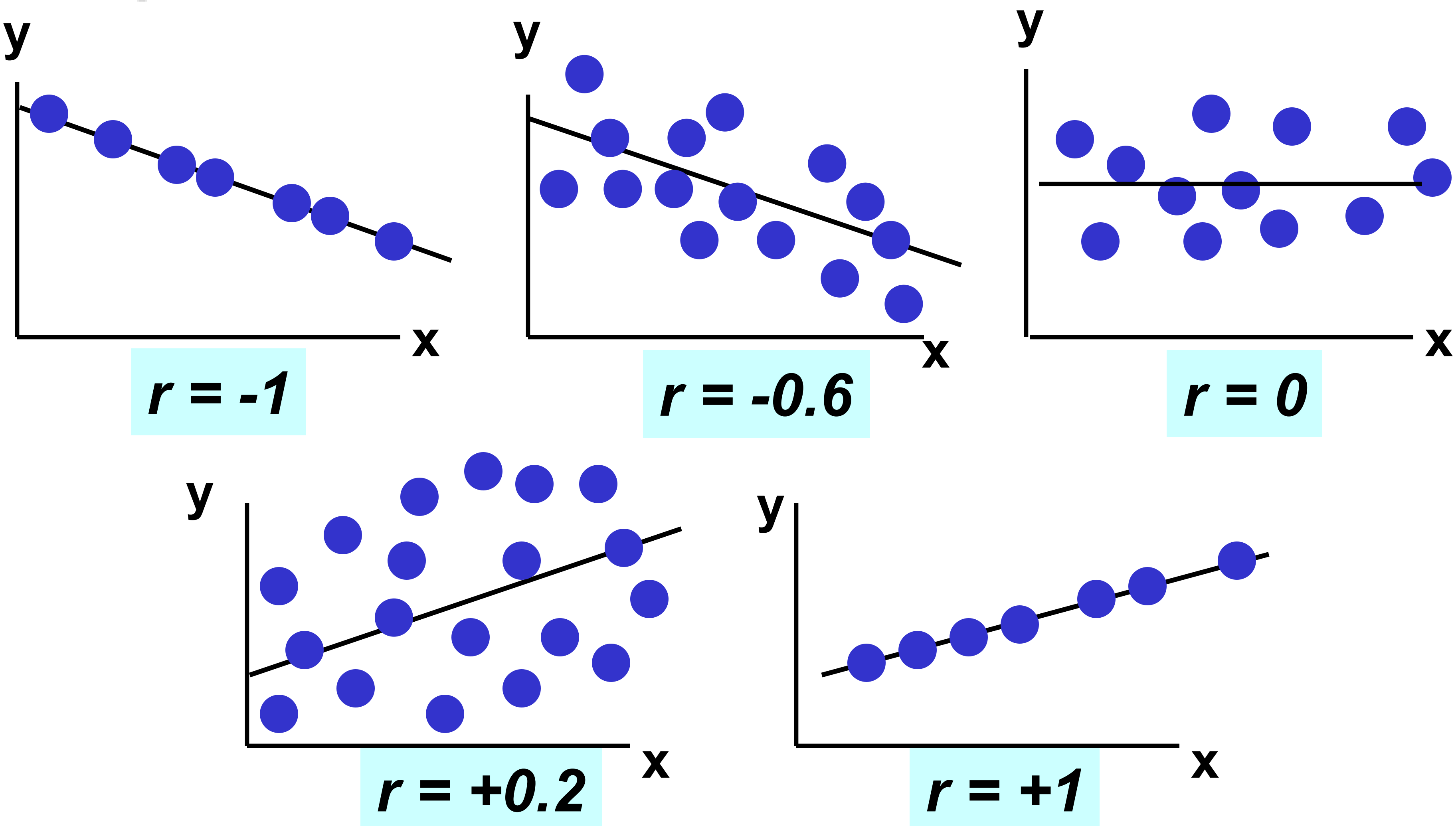
Correlation Coefficient

- The population correlation coefficient ρ (rho) measures the strength of the association between the variables
- The sample correlation coefficient r is an estimate of ρ and is used to measure the strength of the linear relationship in the sample observations

Features of ρ and r

- Unit free
- Range between -1 and 1
- The closer to -1, the stronger the negative linear relationship
- The closer to 1, the stronger the positive linear relationship
- The closer to 0, the weaker the linear relationship

Examples of Approximate *r* Values



Calculating the Correlation Coefficient

Sample correlation coefficient:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2][\sum (y - \bar{y})^2]}}$$

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$$r = \hat{\rho} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{S_X}\right)\left(\frac{Y_i - \bar{Y}}{S_Y}\right) = \frac{S_{XY}}{S_X S_Y}$$

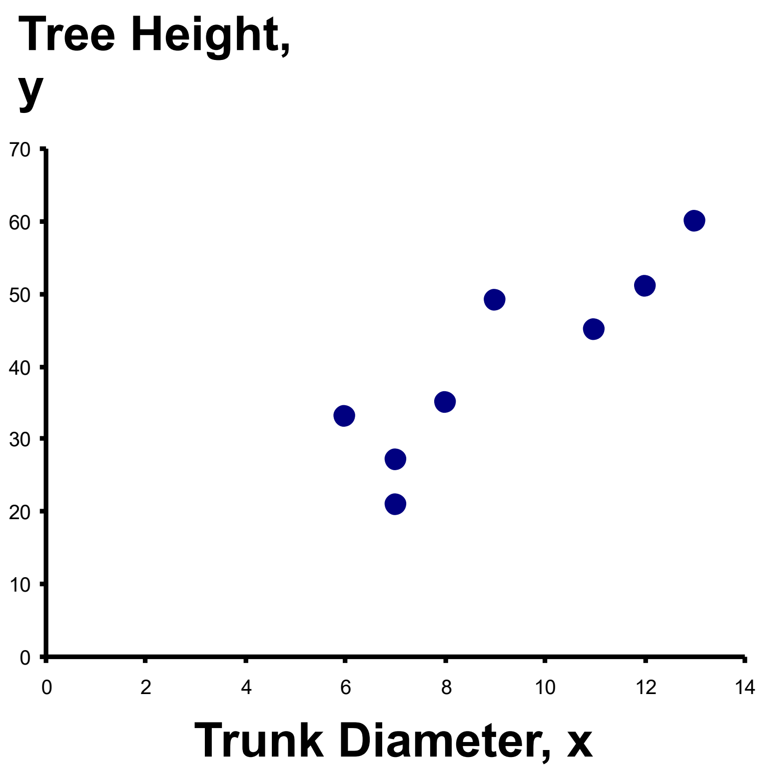
where:

r = Sample correlation coefficient
n = Sample size
x = Value of the independent variable
y = Value of the dependent variable

Calculation Example

Tree Height	Trunk Diameter			
y	x	xy	y^2	x^2
35	8	280	1225	64
49	9	441	2401	81
27	7	189	729	49
33	6	198	1089	36
60	13	780	3600	169
21	7	147	441	49
45	11	495	2025	121
51	12	612	2601	144
Σy =321	Σx =73	Σxy =3142	Σy^2 =14111	Σx^2 =713

Calculation Example



$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

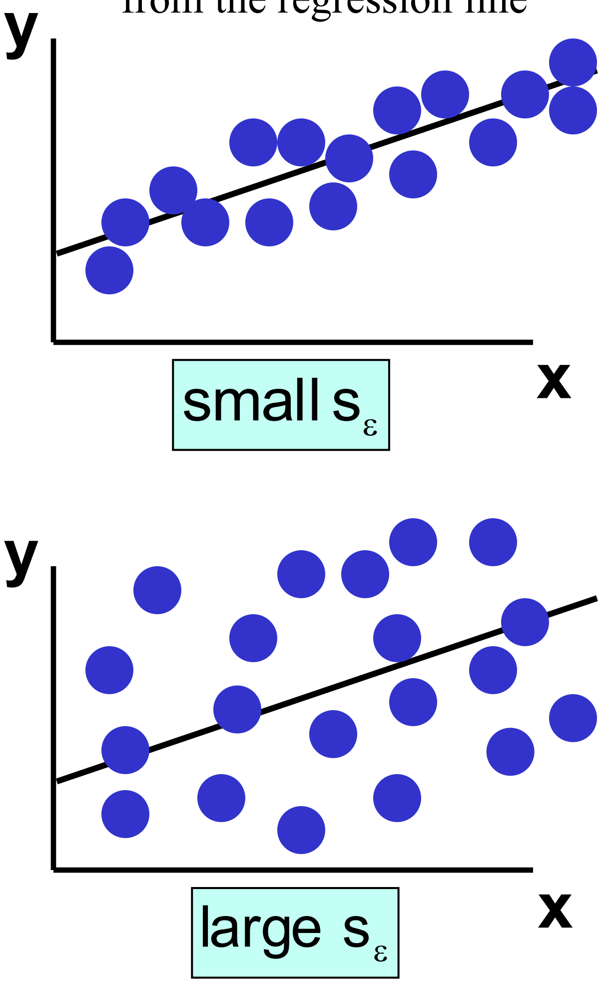
$$= \frac{8(3142) - (73)(321)}{\sqrt{[8(713) - (73)^2][8(14111) - (321)^2]}}$$

$$= 0.886$$

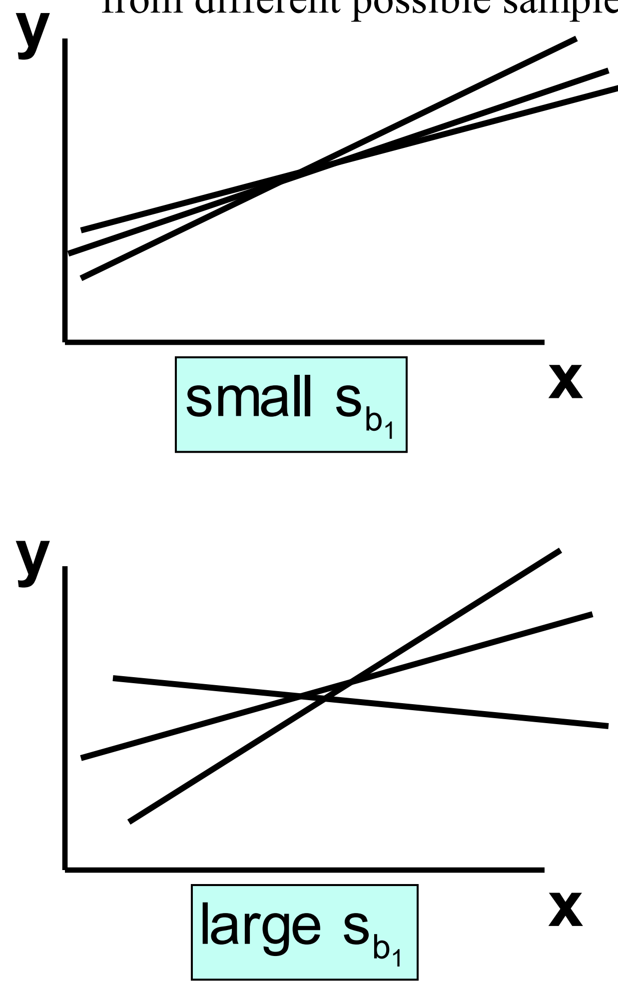
***r* = 0.886** *→ relatively strong positive linear association between x and y*

Comparing Standard Errors

Variation of observed y values from the regression line



Variation in the slope of regression lines from different possible samples



Coefficient of Determination, R^2

- The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable

- The coefficient of determination is also called R-squared and is denoted as R^2

$$\boxed{R^2 = \frac{SSR}{SST}} \quad \text{where} \quad \boxed{0 \le R^2 \le 1}$$

Coefficient of Determination, R^2

$$\boxed{R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\text{sum of squares explained by regression}}{\text{total sum of squares}}}$$

Note: In the **single independent variable** case, the coefficient of determination is

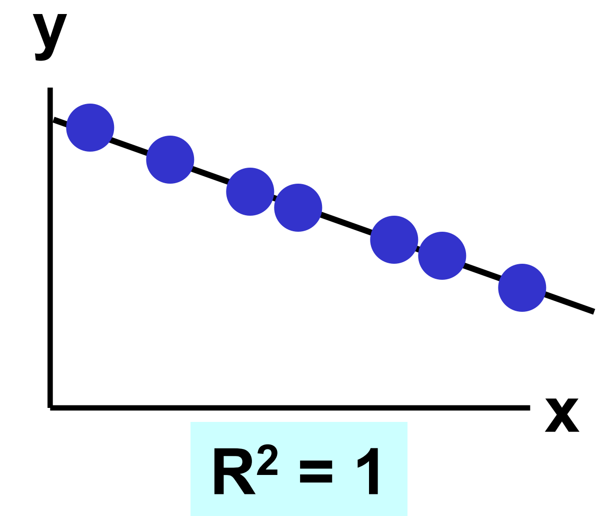
$$\boxed{R^2 = r^2}$$

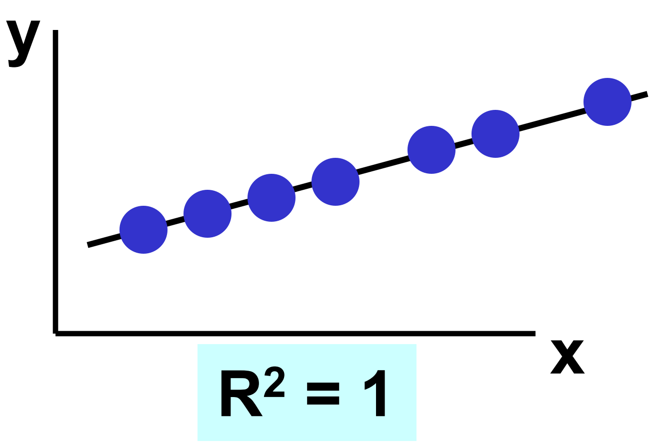
where:

R^2 = Coefficient of determination

r = Simple correlation coefficient

Examples of Approximate R^2 Values



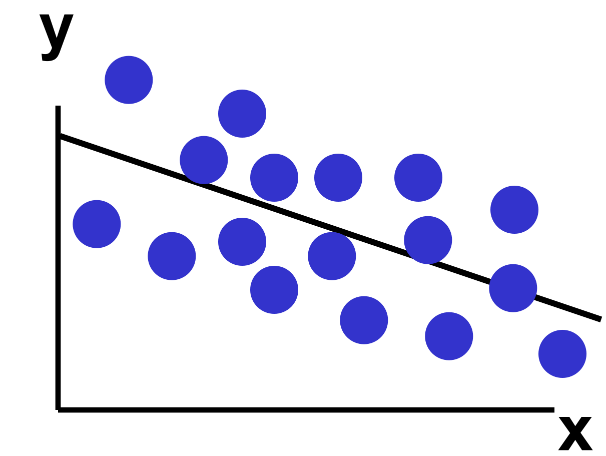


$R^2 = 1$

Perfect linear relationship between x and y:

100% of the variation in y is explained by variation in x

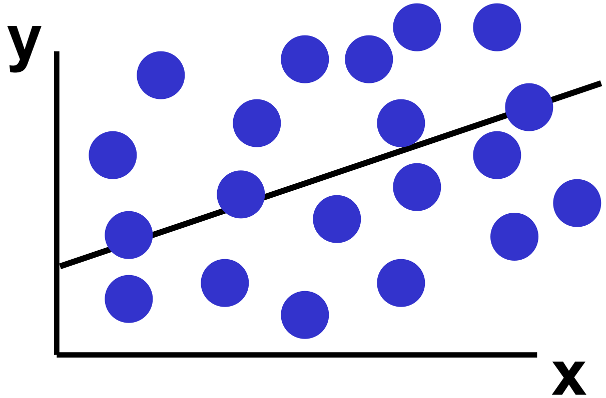
Examples of Approximate R^2 Values



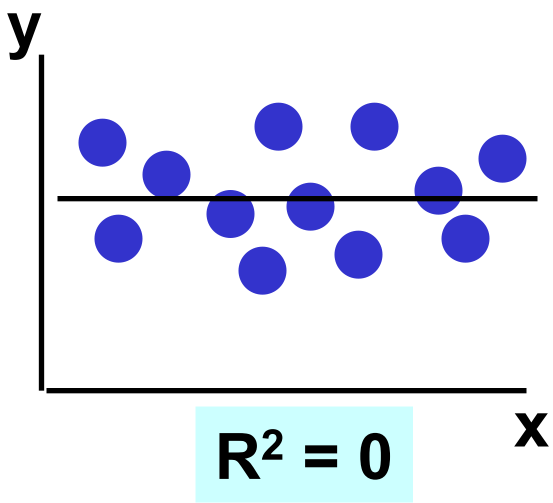
$0 < R^2 < 1$

Weaker linear relationship between x and y:

Some but not all of the variation in y is explained by variation in x



Examples of Approximate R^2 Values



$R^2 = 0$

No linear relationship between x and y:

The value of Y does not depend on x. (None of the variation in y is explained by variation in x)

Thank

You

谢谢